

The emerging role of GATA transcription factors in development and disease.

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The GATA family of transcription factors consists of six proteins (GATA1-6) which are involved in a variety of physiological and pathological processes. GATA1/2/3 are required for differentiation of mesoderm and ectoderm-derived tissues, including the haematopoietic and central nervous system. GATA4/5/6 are implicated in development and differentiation of endoderm- and mesoderm-derived tissues such as induction of differentiation of embryonic stem cells, cardiovascular embryogenesis and guidance of epithelial cell differentiation in the adult.

Expression and genetic interaction of transcription factors GATA-2 and GATA-3 during development of the mouse central nervous system.

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Here we examine the expression of transcription factors GATA-2 and GATA-3 during early stages of embryonic development in the central nervous system (CNS) of the mouse. GATA-2 is expressed as early as 9 dpc in the hindbrain, in ventral rhombomere 4, and transiently in ventral rhombomere 2 (r2). From 9.5 to 11.5 dpc, activation of the gene spreads to many sites of early neuronal differentiation, such as the olfactory bulbs, the pretectum, and the oculomotor nucleus in the midbrain, a thin stripe of cells lining the floor plate from the mesencephalon to the cervical spinal cord and a ventral column of cells spanning the neural tube from rostral hindbrain and including motor neuron as well as ventral interneuron precursors. GATA-3 is expressed in a pattern very similar to that of GATA-2. Distinguishing features are the lack of expression in r2 at 9 dpc and a slight delay in its activation. In addition, GATA-2 is activated in both the ventricular and the subventricular zones of the neural tube, whereas GATA-3 is restricted mainly to the subventricular zone. Expression analyses performed on GATA-2 ^{-/-} mouse embryos between E9.5 and 10.5 dpc established that: (i) the expression of GATA-3 in the developing CNS of the mouse embryo is dependent on the presence of GATA-2 and (ii) loss of GATA-2 leads to severe defects in neurogenesis, which strongly suggests that GATA-2 is involved, as in hematopoiesis, in the maintenance of the pool of ventral neuronal progenitors.

Expression of the transcription factor GATA3 in the postnatal mouse central nervous system.

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GATA binding protein 3 (GATA3) is an important regulator of central nervous system (CNS) development, but its expression pattern in the postnatal CNS has not been studied. In the present study, we examined the distribution of GATA3 mRNA in the mouse CNS at different postnatal stages by in situ hybridization. During the first 2 weeks of postnatal development, numerous GATA3-expressing cells were found in the intergeniculate leaf, ventral lateral geniculate nucleus, pretectal nucleus, nucleus of the posterior commissure, superior colliculus, inferior colliculus, periaqueductal grey, substantia nigra and raphe nuclei. Few notable changes in the profile of GATA3 expression occurred over this time period. As postnatal development progressed, however, GATA3 expression weakened, and was maintained in only a few regions of the adult CNS. Throughout the brain, we found that GATA3-expressing cells were

NeuN-positive, and no colocalization with glial fibrillary acidic protein (GFAP) was observed. In the substantia nigra, GATA3 was exclusively expressed in cells of the reticulate part and some of which were found to be GABAergic. This study presents a comprehensive overview of GATA3 expression in the CNS throughout postnatal life, and the dynamics that we observed provide insights for further investigations of the roles of GATA3 in postnatal development and the maintenance of the mature CNS.

SCL, GATA-2 and Lmo2 expression in neurogenesis.

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SCL, Lmo2 and GATA factors form common transcription complexes during hematopoietic differentiation. The overlapping expression of SCL with GATA-2 and GATA-3 in the developing brain indicated that these factors might collaborate also in the course of neural tissue differentiation. The expression pattern of Lmo2 in the developing CNS, however, is not well understood. Here, we show that neural cells in the early embryonic chick mid- and hindbrain express SCL and GATA-2, while Lmo2 is expressed only in vascular elements. The lack of Lmo2 transcripts in neural cells demonstrated that SCL and GATA-2 cannot form common complexes with Lmo2 in the developing brain. In the course of neural tissue genesis, GATA-2 mRNA appeared prior to the SCL transcript. While GATA-2 expression decreased with maturation, SCL expression persisted at a high level also in post-neurogenic periods. The temporal pattern of SCL and GATA-2/3 expression was investigated also in vitro, in the course of induced neurogenesis by NE-4C neural stem cells. While GATA-2 expression increased from the very beginning of differentiation, SCL expression appeared only in more differentiated cells expressing proneural genes. GATA-3 expression, on the other hand, was detected only in advanced stages of the neuronal maturation, which were characterised by the activation of the Math2 neuronal gene. Similarly to the hematopoietic differentiation, GATA-2 expression precedes the activation of both SCL and GATA-3, and may play roles in the activation of the SCL gene in neuronal development. In contrast to hematopoietic differentiation, however, our results failed to demonstrate co-assembling of GATA factors or SCL with Lmo2. While overlapping expression of GATA-2/3 and SCL was detected, Lmo2 activation could not be demonstrated in neural cells in the investigated period of neuronal development.

Role of the pioneer transcription factor GATA2 in health and disease.

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The transcription factor GATA2 is involved in human diseases ranging from hematopoietic disorders, to cancer, to infectious diseases. GATA2 is one of six GATA-family transcription factors that act as pioneering transcription factors which facilitate the opening of heterochromatin and the subsequent binding of other transcription factors to induce gene expression from previously inaccessible regions of the genome. Although GATA2 is essential for hematopoiesis and lymphangiogenesis, it is also expressed in other tissues such as the lung, prostate gland, gastrointestinal tract, central nervous system, placenta, fetal liver, and fetal heart. Gene or transcriptional abnormalities of GATA2 causes or predisposes patients to several diseases including the hematological cancers acute myeloid leukemia and acute lymphoblastic leukemia, the primary immunodeficiency MonoMAC syndrome, and to cancers of the lung, prostate, uterus, kidney, breast, gastric tract, and ovaries. Recent data has also linked GATA2 expression and mutations to responses to infectious diseases including SARS-CoV-2 and Pneumocystis carinii pneumonia, and to inflammatory disorders such as atherosclerosis. In this article we review the role of GATA2 in the

etiology and progression of these various diseases.

GATA2-dependent and region-specific regulation of Gata2 transcription in the mouse midbrain.

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Transcription factor GATA2 is expressed in numerous mammalian tissues, including neural, hematopoietic, cardiovascular and urogenital systems, and yet it plays important roles in the regulation of tissue-restricted gene expression. The Gata2 gene itself is also under stringent tissue-specific control and multiple cis-regulatory domains have been identified in the Gata2 locus. In this study we sought out and then examined in detail the domains that regulate Gata2 in the midbrain. We identified two discrete domains in the Gata2 promoter that direct midbrain expression; these distal 5H and proximal 2H regulatory domains are located 3.0 and 1.9 kbp, respectively, upstream of the transcriptional initiation site. Importantly, both domains contain GATA factor binding sites. Our analyses further revealed that GATA2 is essential for Gata2 gene expression in the midbrain, whereas GATA3 is not. Both the 2H and 5H domains have the independent ability to activate Gata2 gene expression in the midbrain superior colliculus, whereas the distal-5H domain is additionally capable of activating Gata2 transcription in the inferior colliculus. These results demonstrate that two distinct regulatory domains contribute to the Gata2 gene expression in the mouse midbrain and that Gata2 midbrain transcription is under positive autoregulation.

Human GATA-3 trans-activation, DNA-binding, and nuclear localization activities are organized into distinct structural domains.

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GATA-3 is a zinc finger transcription factor which is expressed in a highly restricted and strongly conserved tissue distribution pattern in vertebrate organisms, specifically, in a subset of hematopoietic cells, in cells within the central and peripheral nervous systems, in the kidney, and in placental trophoblasts. Tissue-specific cellular genes regulated by GATA-3 have been identified in T lymphocytes and the placenta, while GATA-3-regulated genes in the nervous system and kidney have not yet been defined. We prepared monoclonal antibodies with which we could dissect the biochemical and functional properties of human GATA-3. The results of these experiments show some anticipated phenotypes, for example, the definition of discrete domains required for specific DNA-binding site recognition (amino acids 303 to 348) and trans activation (amino acids 30 to 74). The signaling sequence for nuclear localization of human GATA-3 is a property conferred by sequences within and surrounding the amino finger (amino acids 249 to 311) of the protein, thereby assigning a function to this domain and thus explaining the curious observation that this zinc finger is dispensable for DNA binding by the GATA family of transcription factors.

Promoter analysis in living zebrafish embryos identifies a cis-acting motif required for neuronal expression of GATA-2.

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We have used zebrafish embryos to dissect the promoter activity of a gene with a complex expression pattern during embryogenesis. GATA-2 is a transcription factor required for hematopoiesis and is dynamically expressed in hematopoietic tissues and in the central nervous system. Using constructs containing zebrafish GATA-2 genomic flanking sequences and the green fluorescent protein (GFP) reporter gene, we demonstrate that distinct regulatory domains are required for hematopoietic, enveloping layer (EVL), and neuronal expression of GATA-2. During gastrulation, GFP expression is confined to the ventral ectoderm and lateral mesoderm and is lacking in the dorsal shield. Cells derived from the regions expressing GFP give rise to hematopoietic progenitors, EVL cells, and neurons. Deletion analysis of the 7.3-kb GATA-2 promoter region revealed that a 1.1-kb DNA sequence is critical for expression of GATA-2 in neurons. Fine mapping revealed that a 31-bp region is required for neuron enhancer activity, and mutagenesis showed that the DNA motif CCCTCCT is essential for GATA-2 promoter activity in the central nervous system of zebrafish. Our use of zebrafish embryos can be exploited as a whole animal system for the dissection of any developmentally regulated vertebrate promoter.

Oral Infigratinib Therapy in Children with Achondroplasia.

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Achondroplasia is a genetic skeletal condition that results in disproportionately short stature and medical complications throughout life. Infigratinib is an orally bioavailable FGFR1-3 selective tyrosine kinase inhibitor in development for achondroplasia. In this phase 2 dose-finding study, we evaluated the safety and efficacy of oral infigratinib in children with achondroplasia between the ages of 3 and 11 years. A total of 72 children were enrolled in five sequential cohorts to receive daily infigratinib at doses of 0.016 mg per kilogram of body weight (cohort 1), 0.032 mg per kilogram (cohort 2), 0.064 mg per kilogram (cohort 3), 0.128 mg per kilogram (cohort 4), and 0.25 mg per kilogram (cohort 5) for 6 months, followed by 12 months of extended treatment in which the dose in cohorts 1 and 2 could be escalated to the next ascending level at months 6 and 12. The primary safety outcome was the incidence of adverse events that led to a decrease in the dose or discontinuation of infigratinib. The primary efficacy outcome was the change from baseline in the annualized height velocity. During treatment, all the children had at least one adverse event, most of which were mild or moderate in severity; none resulted in treatment discontinuation. In cohort 5, an increased annualized height velocity was observed, which persisted throughout the duration of the study, with a mean change from baseline at 18 months of 2.50 cm per year (95% confidence interval [CI], 1.22 to 3.79; $P = 0.001$). The mean change from baseline in height z score was 0.54 (95% CI, 0.35 to 0.72) relative to an untreated achondroplasia reference population at 18 months; the mean change from baseline in the upper-to-lower body segment ratio was -0.12 (95% CI, -0.18 to -0.06). The administration of oral infigratinib did not result in any apparent major safety signal and increased the annualized height velocity and z score and decreased the upper-to-lower body segment ratio at 18 months of treatment in cohort 5. (Funded by BridgeBio Pharma; PROPEL2 ClinicalTrials.gov number, NCT04265651.).

Infigratinib in children with achondroplasia: the PROPEL and PROPEL 2 studies.

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Achondroplasia is the most common short-limbed skeletal dysplasia resulting from gain-of-function pathogenic variants in fibroblast growth factor receptor 3 (FGFR3) gene, a negative regulator of endochondral bone formation. Most treatment options are symptomatic, targeting medical complications. Infigratinib is an orally bioavailable, FGFR1-3 selective tyrosine kinase inhibitor being investigated as a direct therapeutic strategy to counteract FGFR3 overactivity in achondroplasia. The main objective of PROPEL is to collect baseline data of children with achondroplasia being considered for future enrollment in interventional studies sponsored by QED Therapeutics. The objectives of PROPEL 2 are to obtain preliminary evidence of safety and efficacy of oral infigratinib in children with achondroplasia, to identify the infigratinib dose to be explored in future studies, and to characterize the pharmacokinetic (PK) profile of infigratinib and major metabolites. PROPEL (NCT04035811) is a prospective, noninterventional clinical study designed to characterize the natural history and collect baseline data of children with achondroplasia over 6-24 months. PROPEL 2 (NCT04265651), a prospective, phase II, open-label study of infigratinib in children with achondroplasia, consists of a dose-escalation, dose-finding, and dose-expansion phase to confirm the selected dose, and a PK substudy. Children aged 3-11 years with achondroplasia who completed ≥ 6 months in PROPEL are eligible for PROPEL 2. Primary endpoints include treatment-emergent adverse events and change from baseline in annualized height velocity. Four cohorts at ascending dose levels are planned for dose escalation. The selected dose will be confirmed in the dose-expansion phase. PROPEL and PROPEL 2 are being conducted in accordance with the International Conference on Harmonization Good Clinical Practice guidelines, principles of the Declaration of Helsinki, and relevant human clinical research and data privacy regulations. Protocols have been approved by local health authorities, ethics committees, and institutions as applicable. Parents/legally authorized representatives are required to provide signed informed consent; signed informed assent by the child is also required, where applicable. PROPEL and PROPEL 2 will provide preliminary evidence of the safety and efficacy of infigratinib as precision treatment of children with achondroplasia and will inform the design of future studies of FGFR-targeted agents in achondroplasia. ClinicalTrials.gov: NCT04035811; NCT04265651.